

Microeconomic Analysis: Problem Set V

Due to week 6: submit problems 1, 4 and 6

1 Tutorial Work Preferences

A problem set of Microeconomic Analysis has n exercises. One of the students aims for a tutorial work involving less exercises, but two assignments with a difference of exactly one exercise cannot be distinguished. Therefore, the *preferences* of this student can be described by

$$\text{For every } x, y \in \{0, 1, \dots, n\} : x \succeq y \Leftrightarrow x \leq y + 1$$

- (a) Determine whether these preferences satisfy reflexivity, connectedness, completeness, transitivity, symmetry and antisymmetry.
- (b) Conclude whether these preferences are rational or not.
- (c) Are these preferences representable by a utility function?

2 Utility Representation

Consider the space of bundles $\mathbb{R}_{++}^2$, and the rational preference

$$(x_1, x_2) \succeq (y_1, y_2) \Leftrightarrow x_1^3 x_2 \geq y_1^3 y_2.$$

- (a) Are these preferences representable by a continuous utility function?

- (b) Discuss whether or not $v(x_1, x_2) = x_1^{\frac{3}{4}} x_2^{\frac{1}{4}}$, $w(x_1, x_2) = 3 \log x_1 + \log x_2$, $\tau(x_1, x_2) = \frac{1}{x_1^3 x_2}$ are utility representations of these preferences.
- (c) Construct the *uniform consumption equivalent* utility representation in which the utility of (x_1, x_2) is the amount x such that the bundle (x, x) is indifferent to (x_1, x_2) .

3 College Meals and Revealed Preference

Your college has four meals on the dinner menu: M(eat), F(ish), V(eggies) and P(asta). Depending on when you arrive to the hall, some of them may not be available. Suppose that your Economics tutor asks you to obtain your friend's rational (and stable) strict preferences by observing their choice behaviour over a week in the hall. From Monday to Wednesday, menus were $\{F, M, P\}$, $\{P, V\}$ and $\{M, V\}$.

- (a) Wednesday night, you write an e-mail to your tutor that says: No need to come back tomorrow, job finished! What did the student choose for lunch from Monday to Wednesday? (Find all solutions)
- (b) Explain what was revealed directly and what was revealed indirectly in part (a).
- (c) Suppose that, instead, the observed choices were P, V and V over the three days. Since you want to finish your job by Thursday, what menu would you prefer hall to offer next day?
- (d) If student's choices were instead P, V and M , then what would you conclude?
- (e) Suppose that the student's choices were indeed P, V, M , but you learn that meat dishes on Monday are objectively lower quality than meat dishes on any other day. If the menu on Thursday is not missing any dishes, i.e., it is $\{M, F, V, P\}$, then what will the student choose?

4 Choice Overload

In a letter to the New York Times, a shoe salesman wrote:

as a neophyte shoe salesman, I was told never to show customers more than three pairs of shoes. If they saw more, they would not be able to decide on any of them.

Consider a shop with a stock of shoes A, B, C and D and denote by 0 the no-pair-sold situation in which the client leaves the shop without buying any pair of shoes (this option is always available to the client).

- (a) Discuss the form of the choice function of an almost rational client that always chooses according to a preference $\succsim$ over A, B, C, D , except when the four pairs of shoes are offered, as suggested by the quote.
- (b) Demonstrate that the choice function fails to satisfy property α , and the weak axiom of revealed preference. Is this behaviour rational?
- (c) Suppose that the shop makes 5 pounds with any shoe sold. Moreover, the shop makes extra 15 pounds (say, in future sales), if the shoe sold is the best according to the preference $\succsim$ of the client, extra 10 pounds if the shoe is the second best, and extra 5 pounds if it is the third best. The shop makes 0 pounds if no shoe is sold. Suppose that a random client arrives. Argue that, when the client arrives, it is in the interest of the seller, at least weakly, to show exactly three pairs of shoes.
- (d) If half of the clients have the preference $A \succ B \succ C \succ D$, 30 per cent have preferences $B \succ C \succ D \succ A$ and 20 per cent have preferences $D \succ A \succ B \succ C$, which three pairs of shoes would you show and what would be the expected profit per client?

5 Budget Set

Consider the space of bundles $\mathbb{R}_+^2$. A consumer has strictly monotone preferences and spends all income in the two goods considered. We observe the choices made in two situations, when prices for goods are $(p_1, p_2) = (3, 1)$ and when prices for goods are $(1, 3)$.

- (a) Show that observed choices cannot be $(x_1, x_2) = (2, 1)$ in the first case, and $(1, 2)$ in the second. What if the choices are $(2, 1)$ and $(\frac{7}{4}, \frac{7}{4})$?
- (b) Show that observed choices can be $(1, 3)$ in the first case and $(1, 2)$ in the second.
- (c) For bundles in part (b), discuss whether preferences can be strictly convex or not, and whether they can be Cobb-Douglas or not

6 Time Preferences

Consider a rational individual that has the following preferences over a set of timed consumptions (the first component capturing money, and the second the time at which it is paid)

$$(9, 0) > (9, 1) \sim (4, 0) > (9, 2) > (4, 1) > (4, 2) \sim (1, 0) > (1, 1) > (1, 2)$$

- (a) Show that, although the individual is rational, there is no discounted utility that explains these preferences (and hence, that would explain the choices generated by them). What property is not satisfied?
- (b) Fix the utility function over money $u(x) = \sqrt{x}$. Show that you can find a sort of impatience parameter $\gamma \geq 0$ such that the utility function $U(x, t) = \frac{1}{1+\gamma t} u(x)$ can represent these preferences. Discuss this representation.